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EXP1-1 Round peg in a square hole

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1 Introduction

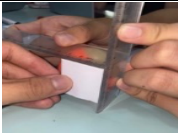
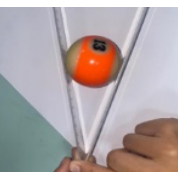
The objective of this experiment involves determining the diameters of three differently-sized balls using the provided tool and then compiling a report on the findings. The primary aim is to enhance our ability to measure things. The challenge lies in measuring the diameter of a non-flat spherical object with the provided tool.

In scientific measurement, precision reflects the consistency and reproducibility of measurements, while accuracy indicates how closely those measurements align with the actual value being measured. In the experiment, errors are categorized into three types. Systematic error is a consistent and predictable deviation in the measurement process; Random error is the measurement variation triggered by various unpredictable factors, which randomly disperses the measurement results near the true value, while measurement error is the total difference between the measured value and the true value, including systematic and random errors, reflecting the accuracy and precision of the measurement.

This article is written with a focus on an experimental measurement.

2 Methods

Table1. Methods for measurements.

No	Method	Schematic	Billiard	Table tennis	Glass ball	Description	Advantages
1.1	Cylindrical method (indirect)		✓	✓		a) Cut paper, wrap around ball, mark ends. b) Measure marks for perimeter, find diameter.	Comparison of 1.2 and 3.2 for medium sized balls to reduce errors.
2	Vertical clamp method		✓	✓		a) Place the two rulers vertically, flush with the scale. b) Put a right angle for measurement.	Straightedge vertical for accurate measurements.
3.1	Ruler intersection method		✓	✓		a) Marking the position of the centre of the ball b) Scale and draw a line c) Make a perpendicular to get the radius d) $d=2r$, calculate the diameter	Compared to method 3.2, no need to ensure the rulers are parallel.
1.2	Cylindrical method (direct)				✓	a) Wrap the ball in a paper tape. b) Measure the diameter directly with a ruler.	Direct measurement, reduced error propagation, suitable for small to medium sized spheres.
3.2	Ruler parallel method				✓	a) Make cardboard b) Make sure the ruler is against its cardboard c) Mark the position for reading	Easier and more precise handling of small balls than Methods 1-2.

3 Results

3.1 Billiard

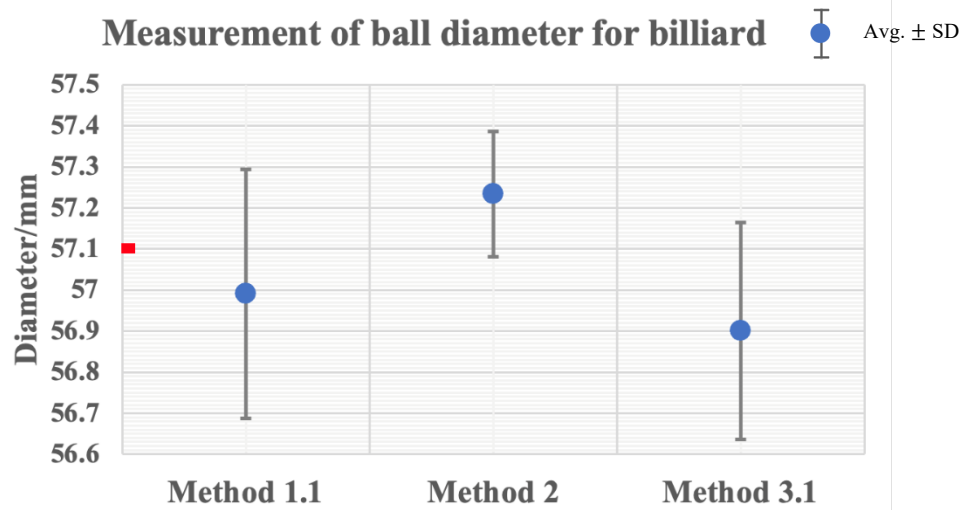


Figure 1. Measurement of ball diameter for billiard by 3 different methods. Data points show the average value of each method, error bars are 1 SD, n=4. The red bar shows the specification of the billiard, 57.1 ± 0.01 mm.

Equation (1) is used to calculate the SD, where the data range of X is from the maximum to the minimum value for a particular method. Equation (2) is used to calculate the absolute error, is the specification of each ball.

$$SD = \sqrt{\frac{\sum (X - \bar{X})^2}{n-1}} \quad (1)$$

$$\text{Absolute error} = |\bar{X} - X^*| \quad (2)$$

Table 2. Billiard measurement Data.

Name of method	Measurement averages	Standard deviation (SD)	Avg. $\pm$ SD	Absolute error
Method 1.1	56.99mm	0.304mm	56.99 $\pm$ 0.304mm	0.11mm
Method 2	57.23mm	0.153mm	57.23 $\pm$ 0.153mm	0.13mm
Method 3.1	56.90mm	0.265mm	56.90 $\pm$ 0.265mm	0.20mm

3.2 Table tennis

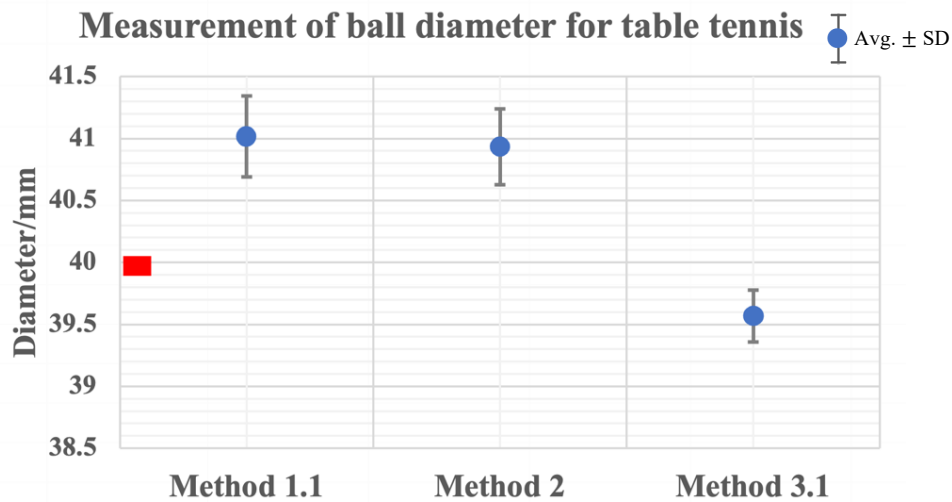


Figure 2. Measurement of ball diameter for tennis ball by 3 different methods. Data points show the average value of each method, error bars are 1 SD, n=4. The red bar shows the specification of table tennis, 40.0 ± 0.1 mm.

Table 3. Table tennis measurement data.

Name of method	Measurement averages	Standard deviation (SD)	Avg. $\pm$ SD	Absolute error
Method 1.1	41.02mm	0.328mm	40.78 $\pm$ 0.328mm	0.78mm
Method 2	40.93mm	0.306mm	40.93 $\pm$ 0.306mm	0.93mm
Method 3.1	39.57mm	0.208mm	39.57 $\pm$ 0.208mm	0.43mm

3.3 Glass Ball

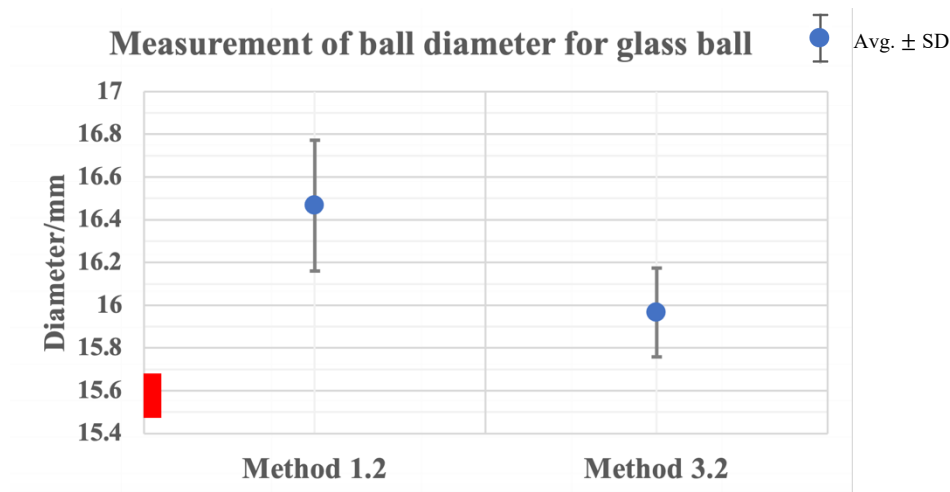


Figure 3. Measurement of ball diameter for glass ball by 2 different methods. Data points show the average value of each method, error bars are 1 SD, n=4. The red bar shows the specification of glass ball, 15.5-15.6mm.

Table 4. Glass ball measurement data.

Name of method	Measurement averages	Standard deviation (SD)	Avg. $\pm$ SD	Absolute error
Method 1.2	16.47mm	0.306mm	16.47 $\pm$ 0.306mm	0.92mm
Method 3.2	15.97mm	0.208mm	15.97 $\pm$ 0.208mm	0.42mm

4 Discussion

4.1 Precision and accuracy

Table 5. Precision and accuracy of each method

	Precise	Accurate
Billiard	Method 2	Method 1.1
Table tennis	Method 3.1	
Glass ball	Method 3.2	

A smaller standard deviation means that the method is more precise, and a smaller absolute error means that the method is more accurate.

4.2 Choice of method

Table 6. Methods and the size, surface of the ball

	Size	Surface	Reasons for choosing the corresponding methods
Billiard	Medium	Slightly rough	Method 1.1: Rough surface for better paper tape winding Method 2: Medium size for easy reading
Table tennis	Medium	Rough	Method 3.1: When using a ruler to hold a ball, the ruler is held at a larger angle to facilitate reading
Glass ball	Small	Smooth	Method 1.2: Small size for easy diameter measurement directly through the paper wall Method 3.2: The small size makes it easier to keep the two rulers parallel.

4.3 Error analysis

4.3.1 Random error

Paper elongation error: When wrapping around a sphere, the paper may elongate due to the force applied to it, resulting in large or small measurements (if within elastic limits, the paper returns to its original length after the force is applied).

Ruler deformation error: When fixing the sphere, the ruler may deform slightly.

Position error: When placing the ruler and the ball, the ruler may not be perfectly perpendicular or parallel.

Human error: Human error occurs when manually marking, measuring, and reading.

4.3.2 Systematic error

Inaccurate ruler scale, thick marker, defective sphere surface.

5 Reflection

This experiment deepened our grasp of indirect measurement and improved our lateral thinking. In engineering, if direct solutions are unattainable, such lateral approaches prove invaluable.

To ensure precision and accuracy, it's crucial to minimize both systematic and random errors. Utilizing precise tools and simplifying steps can aid in this.

Our team excelled in task distribution, communication, and collaboration, leading to the experiment's success.

If revisited, I'd take more measurements for statistical robustness, employ aids like gridlines for accuracy, and refine our method to counteract potential paper elongation and ruler deformation.